

CLAIMS

1. Compositions containing:
 - a) lipophilic extract of *Zingiber officinale* obtained by extraction with
5 carbon dioxide under supercritical conditions; and
 - b) extract of *Cynara scolymus*.
2. Compositions as claimed in claim 1, containing the two components within the following weight ranges:
 - a) lipophilic extract of *Zingiber officinale*: 1 to 25 mg; and
 - 10 b) extract of *Cynara scolymus*: 50 to 200 mg.
3. Compositions as claimed in claims 1 and 2, containing the two components in the following weight amounts:
 - a) lipophilic extract of *Zingiber officinale*: 12.5 mg; and
 - b) extract of *Cynara scolymus*: 100 mg.
- 15 4. Compositions as claimed in claim 1, 2 or 3 wherein the extract of *Cynara scolymus* is obtained by extraction from the aerial parts of the plant with an alcohol or water-alcohol solvent and fractionation on a resin.
5. Compositions as claimed in claim 1, 2 or 3 in which the extract of *Cynara scolymus* contains not less than 20% of caffeoylquinic acids, not less
20 than 5% of flavonoids and not less than 5% of cynaropicrin.
6. Compositions according to the above claims, for oral administration.
7. Compositions as claimed in claim 6 in the form of soft gelatin capsules or cellulose capsules suitable to contain oily substances.
8. The use of:
 - 25 a) lipophilic extract of *Zingiber officinale* obtained by extraction with carbon dioxide under supercritical conditions; and
 - b) extract of *Cynara scolymus*for the preparation of a composition for the reduction of postprandial

drowsiness.

9. The use of:

a) lipophilic extract of *Zingiber officinale* obtained by extraction with carbon dioxide under supercritical conditions; and

5 b) extract of *Cynara scolymus*

for the preparation of a composition having prokinetic and anti-dyspeptic activities.

COMPOSITIONS COMPRISING A LIPOPHILIC EXTRACT OF ZINGIBER OFFICINALE AND AN EXTRACT OF CYNARA SCOLYMUS, WHICH ARE USEFUL FOR THE PREVENTION AND TREATMENT OF OESOPHAGEAL REFLUX AND IRRITABLE BOWEL SYNDROME

Summary of the invention

The present invention relates to compositions consisting of a combination of a lipophilic extract of *Zingiber officinale* and an extract of *Cynara scolymus*, which are useful ~~for the prevention and treatment of~~
5 ~~oesophageal reflux and irritable bowel syndrome, and/or~~ to reduce postprandial drowsiness.

Prior art

Cancer of the gastrointestinal tract is the second most common form of cancer, and the second-highest cause of death when this area is affected.
10 Oesophagus and stomach cancer are not very common in the industrialised countries, but are extremely lethal.

A number of contributory causes have been identified for these two tumours, such as abuse of spirits, often combined with smoking, a nitrate-rich diet, and/or products or habits that break the mucus barrier in the stomach and
15 oesophagus. Following examination of eating habits and deficiencies of some trace elements in populations from different continents (Plummer-Vinson and Paterson-Kelly syndromes), attempts have been made to prevent the onset of carcinoma of the stomach and oesophagus due to said contributory causes by modifying eating habits and lifestyles, and using medicinal or diet products.
20 Although many of said contributory causes have been eliminated, the global incidence has increased because of an increase in the number of people suffering from gastro-oesophageal reflux, often associated with diet and/or

hiatus hernia, which is one of the main causes of oesophageal tumours. Adenocarcinoma is manifested by dysplastic columnar epithelium in the distal part of the oesophagus, nearly always in the presence of chronic reflux and gastric metaplasia of the epithelium (Barrett's oesophagus), which are
5 common in obese individuals, and becomes malignant in over 50% of cases.

It is therefore very important to reduce the incidence of this disorder.

Research is consequently oriented towards drugs able to reduce reflux, especially by modifying gastric voiding. In particular, researchers are attempting to develop well-tolerated natural prokinetics, as the potential drug
10 will have to be taken chronically for many years on a preventive basis, or discontinuously, depending on digestive requirements.

However, many drugs which have been used recently to modify gastric voiding present serious side effects affecting the CNS or have a carcinogenic action, and have therefore either been, or are about to be, withdrawn from the
15 market. Gastro-oesophageal reflux is currently treated with a cocktail of drugs such as proton pump inhibitors (ranitidine, -prazoles or simple antacids), enzymes, various digestive drugs, and prokinetics such as domperidone. However, long-term use of proton pump inhibitors often leads to major digestive problems.

20 Research is consequently now oriented towards new therapeutic approaches.

Irritable bowel syndrome affects up to 9% of the population in industrialised countries. It is a psychosomatic disorder, and is treated with antidepressants, anti-inflammatories, probiotics and other drugs, depending on
25 its severity and duration. Irritable bowel syndrome is mainly exacerbated by stress, inadequate diet, and inflammatory states of various origins. It is also influenced by the state of the stomach. The presence of oesophageal reflux often worsens the conditions associated with irritable bowel syndrome.

<please insert page 2a>

WO 2008/070783 A2, D2 DE 198 59 499 A1 and WO 2008/074080 A1 disclose ginger extract preparations useful for treating dyspepsia, travel sickness, nausea, vomiting and irritable bowel syndrome. The extracts are obtained by supercritical carbon dioxide extraction.

- 5 PHYTOTHERAPY RESEARCH : PTR FEB 2001, vol. 15, no. 1, February 2001 (2001-02), pages 58-61, WO 2008/107183 A1, WO 2007/006391 A2 and WO 03/013562 A1 disclose preparations containing *Cynara scolymus* hydroalcoholic extracts purified on an adsorption resin column and their use in the treatment of irritable bowel syndrome and dyspepsia.

- 10 Brhat Nighantu Ratnakara - Compiled by Gangavisnu Srikrnsna Dasa, Translated by Sri Dattarama Srikrnsnalala Mathura; Vol. 2 (Part IV), edn. 1995, Khemaraja Srikrnsnadas Prakasana, Mumbai 4, pag. 181, discloses a formulation containing *Zingiber officinale Roscoe* (garden ginger, Ginger) as the only active ingredient useful in the treatment of dyspepsia/loss of appetite.

- 15 Agasthiyar; Agasthiar vaithiam 600. Pub.&Ed. Anonymous, pag. 104, 2 discloses a formulation containing *Zingiber officinale Roscoe* (garden ginger, Ginger) as the only active ingredient useful in the treatment of hyperacidity.

- Smkaradajisastrapade; Aryabhisaka - Gujarati Edited (Hindustana No Vaidyaraja) Translation by Harikrishna Bhagwan Lal Vyas; Sastu Sahitya
20 Vardhaka Karyalaya, Bhadra, Ahmedabad, Edn. 12th, 1957, pag. 14, discloses a formulation containing *Zingiber officinale Roscoe* (garden ginger, Ginger) along with *Trichosanthes dioica Roxb.* (pointed gourd) useful in the treatment of hyperacidity and vomiting/emesis.

- Smkaradajisastrapade; Aryabhisaka - Gujarati Edited (Hindustana No Vaidyaraja) Translation by Harikrishna Bhagwan Lal Vyas; Sastu Sahitya
25 Vardhaka Karyalaya, Bhadra, Ahmedabad, Edn. 12th, 1957, pag. 14, discloses a formulation containing *Zingiber officinale Roscoe* (garden ginger, Ginger) along with *Coriandrum sativum Linn.* (Chinese parsley, Chinese-parsley, coriander)

useful in the treatment of hyperacidity.

- 5 Madhavah; Vrindamadhava; Marathi translated by Datto vallala Borkar; Yagyeshwara Gopal Dixit, Bookseller, Pune; Edn. 1922, pag. 519, discloses a formulation containing *Zingiber officinale Roscoe* (garden ginger, Ginger) along with *Thrichosanthes dioica Roxb.* (pointed gourd) useful in the treatment of hyperacidity.

- 10 Yogaratnakarah - Commentary by Lakshmipatisastri, Edited by Brahmasankara Sastri, Chaukhamba Sanskrit Sansthan (Varanasi) Ed. 7th, 2002, pag. 268, discloses a formulation containing *Zingiber officinale Roscoe* (garden ginger, Ginger) along with *Coriandrum sativum Linn.* (Chinese parsley, Chinese-parsley, coriander) useful in the treatment of drowsiness.

Description of the invention

It has now been found that a combination comprising a lipophilic extract of *Zingiber officinale*, prepared by extraction with carbon dioxide under supercritical conditions, and an extract of *Cynara scolymus*, performs a surprising prokinetic and anti-dyspeptic activity, accelerating gastric emptying, eliminating flatulence and improving the digestive function. ~~The combination according to the invention is therefore useful for the prevention and treatment of oesophageal reflux and irritable bowel syndrome.~~

The present invention therefore relates to compositions containing:

- 10 a) lipophilic extract of *Zingiber officinale*, and
- b) *Cynara scolymus* extract,

for the increase of gastric emptying, for the prevention and treatment of ~~oesophageal reflux/gastric voiding and irritable bowel syndrome~~. Moreover, the ability to increase gastric voiding significantly reduces post-prandial drowsiness, thus eliminating the problems associated with it.

More specifically, according to the invention, the lipophilic extract of *Zingiber officinale* will be prepared by extraction from the roots and rhizomes of the plant with carbon dioxide under supercritical conditions, powder being extracted from the root in an extractor at pressures of between 230 and 260 bars, preferably 235 bars, and a temperature of between 40 and 60°C, preferably 50°C, for a time ranging between 1 and 10 hours, preferably seven hours; the extract is collected in the condenser and dehydrated in inert gas dissolved in n-hexane or heptane, and concentrated under vacuum at a temperature not exceeding 40°C. Said extract contains approx. 30% gingerols, and can be used directly in the formulations according to the invention. The *Cynara scolymus* extract will be prepared by conventional methods, for example by extraction from the aerial parts of the plant with an alcohol or water-alcohol solvent and optional fractionation on resin.

The roots and rhizomes of ginger (*Zingiber officinale*), treated in various ways, are used, especially in Asia and the Middle East, as spices and in traditional medicine to treat indigestion, flatulence, diarrhoea, coughing and, to a lesser extent, to protect the mucous membranes, against inflammation, to treat urinary incontinence, etc.

The active components present in the lipophilic extract of *Zingiber officinale* mainly consist of gingerols (generally present in concentrations ranging between 10 and 15%), which possess an anti-dyspeptic, anti-nausea and anti-vomiting activity, and are useful for the treatment of motion sickness, belching, indigestion, colic, vomiting, dyspepsia and stomach and colon pain. However, recent clinical trials have demonstrated that the lipophilic extract of *Zingiber officinale* prepared by traditional methods presents low activity in view of the well-known chemical instability of gingerols; the US pharmacopoeia therefore recommends a complete review of the properties attributed to the plant due to the lack of convincing evidence. The gingerols contained in the lipophilic extract of *Zingiber officinale* prepared by traditional methods break down rapidly, giving rise to a series of compounds, such as shogaol and other products of oxidation, which are devoid of efficacy. These conflicting data are partly due to the instability of the active components in the extracts normally used. However, the extract used in the present invention is a lipophilic extract, stabilised and prepared with carbon dioxide under well-defined supercritical conditions. The *Zingiber officinale* extract accelerates gastric emptying, eliminating postprandial nausea by balancing the adverse effect of the former and completing the therapeutic effect.

Artichoke (*Cynara scolymus*) extracts are known for their cholcretic, cholagogic and anti-dyspeptic action, but often cannot be used by patients suffering from irritable bowel syndrome because they worsen the alternation

of constipation and diarrhoea as a result of the choleretic effect; they also slow gastric voiding and the digestive function.

The *Cynara scolymus* extracts which can be advantageously used according to the invention have a content in three classes of substances in a pre-determined ratio: caffeoylquinic acids, flavonoids derived from luteolin, and cynaropicrin. Caffeoylquinic acids exert a choleretic and liver-protecting effect, flavonoids have a blood-lipid-reducing effect associated with cholesterol synthesis, and cynaropicrin possesses an anti-inflammatory action due to interaction with nuclear factor NFkB and TNF- α .

10 The artichoke extract preferably contains not less than 20% caffeoylquinic acids, not less than 5% flavonoids, and not less than 5% cynaropicrin.

~~In the clinical field, it has surprisingly been found that the combination of these two extracts unpredictably reduces both gastro-oesophageal reflux and the intestinal parameters which are indirect contributory causes by accelerating gastric voiding, eliminating flatulence and improving the digestive function.~~

In patients suffering from irritable bowel syndrome with pain and dyspepsia resistant to the combination of probiotics and antidepressants, the symptoms rapidly disappeared after administration of the composition according to the invention, with a definite improvement in the quality of the life. This activity is not attributable to either of the two components taken separately.

In fact, as already stated, the choleretic, cholagogic and anti-dyspeptic action of artichoke extracts often cannot be used in patients with irritable bowel syndrome, because the choleretic effect worsens the alternation of episodes of constipation and diarrhoea, and reduces the rate of gastric voiding and the digestive function.

The present combination surprisingly improves gastric emptying and digestive function, possesses anti-dyspeptic activity, ~~and is useful for the prevention and treatment of gastro-oesophageal reflux and irritable bowel syndrome, Thus theoretically the present combination prevents the onset of~~
5 ~~oesophageal tumours.~~

According to a preferred aspect, the compositions according to the invention will contain the two components within the following weight intervals:

- a) lipophilic extract of *Zingiber officinale*: from 1 to 25 mg; and
- 10 b) extract of *Cynara scolymus*: from 50 to 200 mg.

According to a particularly preferred aspect, the compositions will contain the two components in the following quantities by weight:

- a) lipophilic extract of *Zingiber officinale*: 12.5 mg; and
- b) extract of *Cynara scolymus*: 100 mg.

15 According to a preferred aspect, the lipophilic extract of *Zingiber officinale* and the *Cynara scolymus* extract will be formulated in vegetable oils rich in $\omega 3/\omega 6$ polyunsaturated fatty acids, such as evening primrose oil.

According to a further aspect, the compositions according to the invention may be administered together with other substances having a useful
20 or complementary activity.

The doses which have proved active in man are 1 to 25 mg of lipophilic extract of *Zingiber officinale* and 50 to 200 mg of *Cynara scolymus* extract; more particularly, 100 mg of *Cynara scolymus* extract and 12.5 mg of *Zingiber officinale* extract per dose, to be taken with every main meal or
25 otherwise adapted to the disorder to be treated.

The compositions according to the invention will be formulated according to conventional methods, such as those described in "Remington's Pharmaceutical Handbook", Mack Publishing Co., N.Y., USA. In particular,

the compositions according to the invention will be formulated according to conventional formulation techniques for lipophilic ingredients intended for oral administration, mainly as sublingual tablets or soft gelatin or cellulose capsules for oils designed to disperse rapidly in the stomach. The lipophilic form uses oils rich in ω -3 fatty acids, which promote rapid absorption of the active ingredient, to disperse the active components. Examples of oral formulations are tablets, dragées, soft and hard gelatin capsules, and cellulose capsules.

The examples set out below further illustrate the invention.

10 **Example 1- Extraction of *Cynara scolymus***

Load 1000 g of dried, minced artichoke leaves into a percolator fitted with a heating jacket, and extract 5 times with 4 L of 70% EtOH at 70°C.

Combine the filtered percolates and concentrate under vacuum to a dry residue of approx. 15%. Leave to cool at ambient temperature, then centrifuge.

15 Load the supernatant solution into a column packed with 1.5 L of AMBERLITE™ XAD™ 7HP resin (Rohm and Haas) previously suspended in water.

Wash the column with 1.5 L of water, then elute with 3.75 L of 90% EtOH. Concentrate the water-alcohol eluate until soft, and then dry at 50°C under vacuum for 24 hours: 82.4 g of purified artichoke leaf extract is obtained.

HPLC titres: total caffeoylquinic acids 31.76%, total flavonoids 14.31%, cynaropicrin 17.51%.

Example 2 - Soft gelatine capsules

25 **Unit composition:**

Extract of <i>Cynara scolymus</i>	150 mg
Lipophilic extract of <i>Zingiber officinale</i>	12.5 mg
Glyceryl monostearate	30 mg

Soya lecithin	10 mg
Evening primrose oil q.s. for	700 mg

Example 3 - Hard gelatin capsules

Unit composition:

5	Extract of <i>Cynara scolymus</i>	100 mg
	Lipophilic extract of <i>Zingiber officinale</i>	12.5 mg
	Microcrystalline cellulose	300 mg
	Lactose	170 mg
	Silicon dioxide	5 mg
10	Magnesium stearate	5 mg